

Group Project Assignment – Database Design and SQL Implementation

Nihon Studio

Sistem Pemesanan Fotografi & Videografi

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1. ERD dan Diagram Pemetaan Data

1.1 Gambaran Umum Sistem

Nihon Studio adalah sistem pemesanan layanan fotografi dan videografi berbasis web. Pelanggan dapat melakukan pemesanan secara online dengan memilih lokasi, layanan, serta tanggal dan waktu yang diinginkan. Database dirancang untuk menyimpan data pelanggan, detail pemesanan, pilihan layanan, dan informasi lokasi.

1.2 Entitas dan Atribut

Entitas	Primary Key	Atribut Utama
Customer	customer_id (PK)	full_name, email, phone, created_at
Service	service_id (PK)	service_name, description, base_price
Location	location_id (PK)	location_type, address, maps_link
Addon	addon_id (PK)	addon_name, price
Booking	booking_id (PK)	customer_id (FK), service_id (FK), location_id (FK), addon_id (FK), booking_date, booking_time, group_size, notes, status, total_price, created_at

1.3 Entity Relationship Diagram (ERD)

Berikut adalah representasi ERD dalam notasi teks yang menggambarkan hubungan antar entitas pada sistem Nihon Studio:

CUSTOMER (customer_id PK, full_name, email, phone, created_at)	
1	_____ N
BOOKING (booking_id PK, customer_id FK, service_id FK, location_id FK, addon_id FK, booking_date, booking_time, group_size, notes, status, total_price, created_at)	
— N	_____ 1 SERVICE (service_id PK, service_name, base_price)
— N	_____ 1 LOCATION (location_id PK, location_type, address)
— N	_____ 1 ADDON (addon_id PK, addon_name, price) [nullable]

1.4 Kardinalitas Relasi

Relasi	Kardinalitas	Keterangan	Letak FK
Customer → Booking	1 : N	Satu pelanggan dapat memiliki banyak pemesanan	booking.customer_id
Service → Booking	1 : N	Satu layanan dapat dipesan berkali-kali	booking.service_id
Location → Booking	1 : N	Satu lokasi dapat digunakan di banyak pemesanan	booking.location_id
Addon → Booking	1 : N (nullable)	Tambahan layanan bersifat opsional	booking.addon_id

1.5 Pemetaan Data — Tabel Relasional

Tabel: customer

Kolom	Tipe Data	Key	Null	Keterangan
customer_id	INT	PK, AI	NOT NULL	ID unik pelanggan
full_name	VARCHAR(100)		NOT NULL	Nama lengkap pelanggan
email	VARCHAR(150)	UNIQUE	NOT NULL	Alamat email pelanggan
phone	VARCHAR(20)		NOT NULL	Nomor telepon
created_at	TIMESTAMP		DEFAULT NOW()	Waktu data dibuat

Tabel: service

Kolom	Tipe Data	Key	Null	Keterangan
service_id	INT	PK, AI	NOT NULL	ID unik layanan
service_name	VARCHAR(100)		NOT NULL	Nama layanan
description	TEXT		NULL	Deskripsi layanan
base_price	DECIMAL(10,2)		NOT NULL	Harga dasar layanan (Rupiah)

Tabel: location

Kolom	Tipe Data	Key	Null	Keterangan
location_id	INT	PK, AI	NOT NULL	ID unik lokasi
location_type	ENUM		NOT NULL	'Studio', 'Outdoor', 'Home Service'
address	VARCHAR(255)		NULL	Alamat fisik (untuk outdoor/home)
maps_link	VARCHAR(500)		NULL	Link Google Maps

Tabel: addon

Kolom	Tipe Data	Key	Null	Keterangan
addon_id	INT	PK, AI	NOT NULL	ID unik tambahan layanan
addon_name	VARCHAR(100)		NOT NULL	Nama tambahan layanan
price	DECIMAL(10,2)		NOT NULL	Harga tambahan (Rupiah)

Tabel: booking

Kolom	Tipe Data	Key	Null	Keterangan
booking_id	INT	PK, AI	NOT NULL	ID unik pemesanan
customer_id	INT	FK	NOT NULL	Mengacu ke customer(customer_id)
service_id	INT	FK	NOT NULL	Mengacu ke service(service_id)
location_id	INT	FK	NOT NULL	Mengacu ke location(location_id)
addon_id	INT	FK	NULL	Mengacu ke addon(addon_id) — opsional
booking_date	DATE		NOT NULL	Tanggal sesi yang dipilih
booking_time	TIME		NOT NULL	Jam sesi yang dipilih
group_size	INT		NOT NULL	Jumlah orang dalam sesi
notes	TEXT		NULL	Catatan tambahan dari pelanggan
status	ENUM		NOT NULL	'Pending', 'Confirmed', 'Cancelled'
total_price	DECIMAL(10,2)		NOT NULL	Total harga pemesanan
created_at	TIMESTAMP		DEFAULT NOW()	Waktu pemesanan dikirim

2. Skrip DDL (Data Definition Language)

Berikut adalah skrip SQL yang digunakan untuk membuat struktur database Nihon Studio beserta seluruh tabelnya dengan constraint yang sesuai.

```
-- =====
-- DDL Script: Sistem Pemesanan Nihon Studio
-- DBMS: MySQL
-- =====

-- 1. Membuat dan menggunakan database
CREATE DATABASE IF NOT EXISTS nihon_studio
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

USE nihon_studio;

-- 2. Tabel: customer (pelanggan)
CREATE TABLE customer (
  customer_id INT NOT NULL AUTO_INCREMENT,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE,
  phone VARCHAR(20) NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (customer_id)
);

-- 3. Tabel: service (layanan)
CREATE TABLE service (
  service_id INT NOT NULL AUTO_INCREMENT,
  service_name VARCHAR(100) NOT NULL,
  description TEXT,
  base_price DECIMAL(10,2) NOT NULL,
  PRIMARY KEY (service_id)
);

-- 4. Tabel: location (lokasi sesi)
CREATE TABLE location (
  location_id INT NOT NULL AUTO_INCREMENT,
  location_type ENUM('Studio','Outdoor','Home Service') NOT NULL,
  address VARCHAR(255),
  maps_link VARCHAR(500),
  PRIMARY KEY (location_id)
);

-- 5. Tabel: addon (tambahan layanan)
CREATE TABLE addon (
```

```

    addon_id    INT                NOT NULL AUTO_INCREMENT,
    addon_name  VARCHAR(100)       NOT NULL,
    price       DECIMAL(10,2)      NOT NULL,
    PRIMARY KEY (addon_id)
);

```

-- 6. Tabel: booking (pemesanan) – dibuat terakhir karena memiliki 4 FK

```

CREATE TABLE booking (
    booking_id    INT                NOT NULL AUTO_INCREMENT,
    customer_id   INT                NOT NULL,
    service_id    INT                NOT NULL,
    location_id   INT                NOT NULL,
    addon_id      INT,
    booking_date  DATE                NOT NULL,
    booking_time  TIME                NOT NULL,
    group_size    INT                NOT NULL DEFAULT 1,
    notes         TEXT,
    status        ENUM('Pending','Confirmed','Cancelled') NOT NULL DEFAULT
'Pending',
    total_price   DECIMAL(10,2)      NOT NULL,
    created_at    TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (booking_id),
    CONSTRAINT fk_booking_customer
        FOREIGN KEY (customer_id) REFERENCES customer(customer_id) ON DELETE
CASCADE,
    CONSTRAINT fk_booking_service
        FOREIGN KEY (service_id) REFERENCES service(service_id) ON DELETE
RESTRICT,
    CONSTRAINT fk_booking_location
        FOREIGN KEY (location_id) REFERENCES location(location_id) ON DELETE
RESTRICT,
    CONSTRAINT fk_booking_addon
        FOREIGN KEY (addon_id) REFERENCES addon(addon_id) ON DELETE SET
NULL
);

```

3. Query CRUD (Perintah DML)

3.1 INSERT — Menambahkan Data

Menambahkan data layanan (service)

```
INSERT INTO service (service_name, description, base_price) VALUES
('Selfie',      'Sesi foto selfie personal di studio.',      150000.00),
('Photopass',   'Dokumentasi foto lengkap dengan photopass.', 250000.00),
('Event',       'Fotografi dan videografi acara lengkap.',   500000.00);
```

Menambahkan data lokasi (location)

```
INSERT INTO location (location_type, address, maps_link) VALUES
('Studio',      'Jl. Nihon No. 1, Batam Center, Batam',
'https://maps.google.com/?q=Nihon+Studio'),
('Outdoor',     'Taman Kota Batam, Batam',
'https://maps.google.com/?q=Taman+Kota+Batam'),
('Home Service', NULL, NULL);
```

Menambahkan data tambahan layanan (addon)

```
INSERT INTO addon (addon_name, price) VALUES
('Printed Photos', 50000.00),
('Extra Editing',  75000.00),
('Photo Album',   120000.00);
```

Menambahkan data pelanggan (customer)

```
INSERT INTO customer (full_name, email, phone) VALUES
('Andi Pratama', 'andi@email.com', '081234567890'),
('Sari Dewi',    'sari@email.com', '082345678901'),
('Budi Santoso', 'budi@email.com', '083456789012');
```

Menambahkan data pemesanan (booking)

```
INSERT INTO booking
(customer_id, service_id, location_id, addon_id,
 booking_date, booking_time, group_size, notes, total_price)
VALUES
(1, 1, 1, 1, '2025-06-15', '10:00:00', 2, 'Please use natural
lighting.', 200000.00),
(2, 2, 2, NULL, '2025-06-18', '14:00:00', 4, 'Outdoor session near the
lake.', 250000.00),
(3, 3, 3, 2, '2025-06-20', '09:00:00', 8, 'Corporate event
documentation.', 575000.00);
```

3.2 SELECT — Menampilkan Data

Menampilkan semua pemesanan beserta detail lengkap (JOIN 4 tabel)

```
SELECT
    b.booking_id,
    c.full_name      AS nama_pelanggan,
    c.email,
    c.phone,
    s.service_name   AS layanan,
    l.location_type  AS tipe_lokasi,
    l.address        AS alamat,
    a.addon_name     AS tambahan,
    b.booking_date   AS tanggal,
    b.booking_time   AS jam,
    b.group_size     AS jumlah_orang,
    b.total_price    AS total_harga,
    b.status
FROM booking b
JOIN customer c ON b.customer_id = c.customer_id
JOIN service s ON b.service_id = s.service_id
JOIN location l ON b.location_id = l.location_id
LEFT JOIN addon a ON b.addon_id = a.addon_id
ORDER BY b.created_at DESC;
```

Menampilkan pemesanan dengan status Pending

```
SELECT b.booking_id, c.full_name, b.booking_date, b.status
FROM booking b
JOIN customer c ON b.customer_id = c.customer_id
WHERE b.status = 'Pending'
ORDER BY b.booking_date ASC;
```

Menghitung jumlah pemesanan per layanan

```
SELECT s.service_name AS layanan,
       COUNT(b.booking_id) AS jumlah_pemesanan
FROM service s
LEFT JOIN booking b ON s.service_id = b.service_id
GROUP BY s.service_name
ORDER BY jumlah_pemesanan DESC;
```

3.3 UPDATE — Mengubah Data

Mengkonfirmasi pemesanan (mengubah status menjadi Confirmed)


```
UPDATE booking
SET status = 'Confirmed'
WHERE booking_id = 1;
```

Memperbarui nomor telepon pelanggan

```
UPDATE customer
SET phone = '089876543210'
WHERE customer_id = 1;
```

Memperbarui harga layanan Selfie

```
UPDATE service
SET base_price = 175000.00
WHERE service_name = 'Selfie';
```

3.4 DELETE — Menghapus Data

Menghapus satu data pemesanan

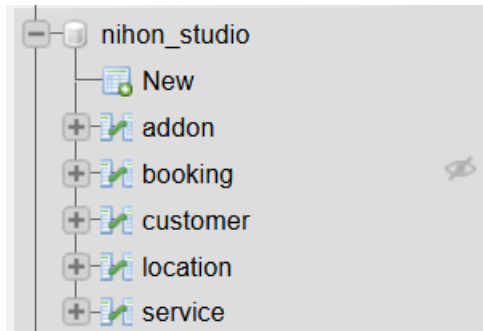
```
DELETE FROM booking
WHERE booking_id = 3;
```

Menghapus data pelanggan beserta seluruh pemesanannya

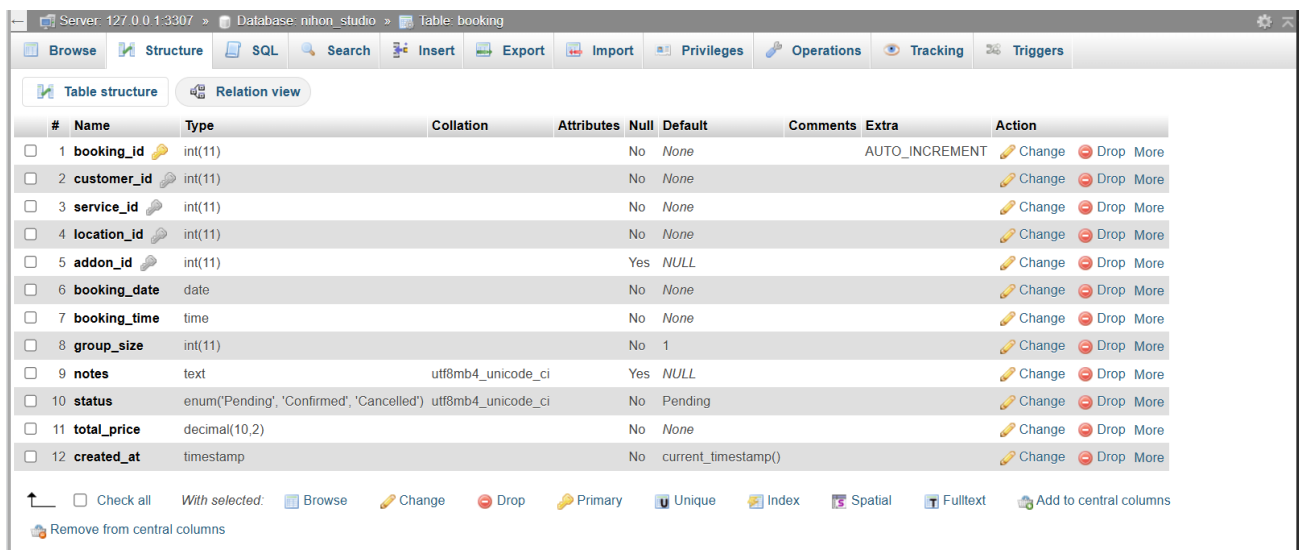
```
-- ON DELETE CASCADE berarti seluruh data booking milik pelanggan ini
-- akan ikut terhapus secara otomatis.
DELETE FROM customer
WHERE customer_id = 3;
```

4. Screenshot Hasil Implementasi

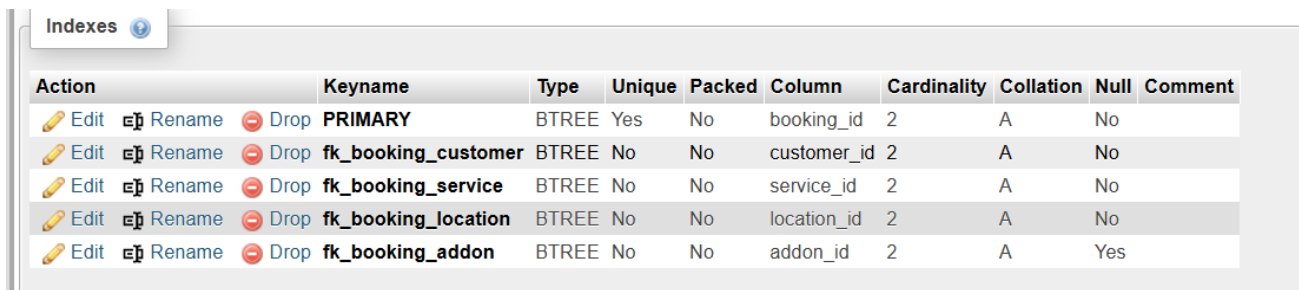
- CREATE DATABASE



- CREATE TABLE



#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	booking_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	customer_id	int(11)			No	None			Change Drop More
3	service_id	int(11)			No	None			Change Drop More
4	location_id	int(11)			No	None			Change Drop More
5	addon_id	int(11)			Yes	NULL			Change Drop More
6	booking_date	date			No	None			Change Drop More
7	booking_time	time			No	None			Change Drop More
8	group_size	int(11)			No	1			Change Drop More
9	notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
10	status	enum('Pending', 'Confirmed', 'Cancelled')	utf8mb4_unicode_ci		No	Pending			Change Drop More
11	total_price	decimal(10,2)			No	None			Change Drop More
12	created_at	timestamp			No	current_timestamp()			Change Drop More



Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	booking_id	2	A	No	
Edit Rename Drop	fk_booking_customer	BTREE	No	No	customer_id	2	A	No	
Edit Rename Drop	fk_booking_service	BTREE	No	No	service_id	2	A	No	
Edit Rename Drop	fk_booking_location	BTREE	No	No	location_id	2	A	No	
Edit Rename Drop	fk_booking_addon	BTREE	No	No	addon_id	2	A	Yes	

- Hasil INSERT

		customer_id	full_name	email	phone	created_at
☐	Edit Copy Delete	1	Andi Pratama	andi@email.com	089876543210	2026-05-22 21:44:00
☐	Edit Copy Delete	2	Sari Dewi	sari@email.com	082345678901	2026-05-22 21:44:00

		service_id	service_name	description	base_price
☐	Edit Copy Delete	1	Selfie	Personal selfie photo session at studio.	175000.00
☐	Edit Copy Delete	2	Photopass	Full event photo documentation with photopass.	250000.00
☐	Edit Copy Delete	3	Event	Complete event photography and videography.	500000.00

		location_id	location_type	address	maps_link
☐	Edit Copy Delete	1	Studio	Jl. Nihon No. 1, Batam Center, Batam	https://maps.google.com/?q=Nihon+Studio+Batam
☐	Edit Copy Delete	2	Outdoor	Taman Kota Batam, Batam	https://maps.google.com/?q=Taman+Kota+Batam
☐	Edit Copy Delete	3	Home Service	NULL	NULL

		addon_id	addon_name	price
☐	Edit Copy Delete	1	Printed Photos	50000.00
☐	Edit Copy Delete	2	Extra Editing	75000.00
☐	Edit Copy Delete	3	Photo Album	120000.00

		booking_id	customer_id	service_id	location_id	addon_id	booking_date	booking_time	group_size	notes	status	total_price	created_at
☐	Edit Copy Delete	1	1	1	1	1	2025-06-15	10 00 00		2 Please use natural lighting.	Confirmed	200000.00	2026-05-22 21:44:00
☐	Edit Copy Delete	2	2	2	2	NULL	2025-06-18	14 00 00		4 Outdoor session near the lake.	Pending	250000.00	2026-05-22 21:44:00

- SELECT dengan JOIN

booking_id	nama_pelanggan	layanan	lokasi	booking_date	status
1	Andi Pratama	Selfie	Studio	2025-06-15	Confirmed
2	Sari Dewi	Photopass	Outdoor	2025-06-18	Pending

- UPDATE status pemesanan

✓ 1 row affected. (Query took 0.0004 seconds.)

```
UPDATE customer SET phone = '089876543210' WHERE customer_id = 2;
```

[Edit inline] [Edit] [Create PHP code]

- DELETE pemesanan

✓ 0 rows deleted. (Query took 0.0003 seconds.)

```
DELETE FROM booking WHERE booking_id = 3;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]